

Latent Travel Motives in the Seoul Metropolitan Area from Hourly Origin–Destination Data

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Abstract

Mobility data derived from telecom records give dense hourly flows between administrative districts, but they do not say *why* people move. We propose a Poisson model that decomposes the Seoul–KT metropolitan mobility data (2024, 481 nodes) into flows generated by a small number of latent travel motives. Each motive has an origin distribution, a destination distribution, a daily rhythm for weekdays and for weekends, and a distance-decay scale. The model is trained on flow counts only; the trip-purpose, nationality and distance labels already present in the data are used only for evaluation. We evaluate six hypotheses, each written down in advance together with an expected outcome and a reason, against held-out labels, against smart-card and road sensors that are independent of the mobility data, on real event days, and on other years. With six motives (6k parameters) the model reaches the same validation deviance as a static OD model with 463k parameters; with eight motives it is 12% lower. Forcing all motives to share one daily rhythm makes the deviance 47% worse, which confirms that motives need their own rhythms. The daily intensity of the commuting motives correlates with same-weekday residuals of smart-card subway entries and exits at 0.66–0.77, and on the days of the Seoul International Fireworks Festival and of rallies in Yeouido the model leaves excess arrivals of +44% to +56% concentrated in Yeouido-dong (ordinary Saturdays: –23% and –32%). In contrast, recovering the held-out purpose labels beats a time-of-day-only predictor by only 1.6 percentage points, and no short-term-foreigner motive separates, because short-term foreigners have essentially the same hourly flow pattern as Korean nationals (correlation 0.976). As a side result, commute distances in Seoul did not increase over 2023–2026 (mean 11.11 km to 10.88 km). We also report the hypotheses that failed and the errors we found and corrected during the analysis.

Keywords: human mobility, origin–destination matrix, latent motives, Poisson factorization, telecom data, external validation

1 Introduction

Mobility data have become abundant, but the reasons for travel remain hard to observe. Household travel surveys ask about purpose directly, but they are run only once every few years and on small samples. Telecom-based mobility data cover the whole city every day, but their trip purposes are values inferred by the provider’s rules. We ask the question from the opposite direction: can we infer the hidden motives that generate travel from the pattern of flows alone, without purpose labels? And do the inferred motives agree with information the model has never seen?

Our approach is as follows. We treat the hourly flow between districts as the sum of a few motives, and give each motive parameters for “where trips tend to start, where they tend to end,

at what hour they tend to happen, and how far they tend to go”. All of these parameters can be read by a person. After training, we test the model against (i) purpose and nationality labels that exist in the data but were not used for training, (ii) smart-card and road-detector records whose measurement has nothing to do with the mobility data, (iii) events with known dates, and (iv) data from other years.

We fixed the evaluation items before training, and wrote an expected outcome and its reason next to each hypothesis (Section 4.3). We report results against these expectations and separate those that held from those that did not, so that we do not report only the hypotheses that worked.

Contributions. (1) A latent-motive decomposition that uses labels for evaluation only, together with an external-validation protocol. (2) A quantitative account, on 2024 data for the Seoul metropolitan area, of where the motives hold and where they do not. (3) A record of structural problems in the data (encoding changes, district-code changes, the behavioral meaning of the labels) and of the errors we found and fixed during the analysis. (4) Side observations from the same data: the trend in commute distance over 2023–2026 and the nature of the short-term-foreigner label.

2 Related work

Human mobility and telecom-based OD. Mobile-phone records showed that individual mobility is highly regular and concentrated on a few places [4]. Estimating OD flows from telecom records was addressed in [3], inferring trip purpose and time of day from telecom data in [1], and a framework that generates mobility behavior without travel surveys was proposed in [6]. Reading the spatial structure of cities from telecom data is studied in [8]. The purpose labels we use are themselves products of such inference, so we treat them as a point of comparison rather than as ground truth.

Spatial interaction and decomposition models. Gravity-type models that explain flows by the size of origin and destination and by distance decay have a long tradition [10, 9]. Non-negative matrix factorization [7] and latent topic models [2] split an observed total into a few parts. Our model combines the two: a Poisson factorization in which each motive has a gravity-type spatial structure and its own daily rhythm.

Relation to our earlier work. In earlier work we compressed Seoul’s de facto population into latent vectors as an urban vibrancy embedding [5]. The present study addresses the movement (flow) side of the same family of data. It changed direction after a preliminary experiment in which an autoencoder embedding of daily patterns did not beat mean profiles (Section 5.10).

3 Data

3.1 Seoul metropolitan mobility

We use the “Seoul metropolitan mobility” data released by the Seoul Metropolitan Government and KT, specifically the trips between origin and destination districts (Seoul Open Data Plaza, OA-22300). Each daily file gives the origin and destination district, departure hour, Korean/foreign status, trip purpose, mean trip distance and duration, and the number of travelers. The data cover 1,339 days from 2023-01-01 to 2026-08-31; the main analysis uses 2024 (366 days). Departure

time is given in 20-minute bins for 7–9 and 17–19 h and in one-hour bins otherwise; we aggregate to hours. The seven purposes are commute to work, commute to school, return home, shopping, tourism, hospital, and other.

Nature of the labels. According to the Seoul mobility guide, purposes are assigned by first estimating each person’s nighttime location (home), daytime location (work or school) and other places from telecom records, and then classifying a trip by the type of movement between them. A trip is counted only if the person stays at least 27 minutes at the endpoint. Work, school and return-home trips are therefore essentially the provider’s rule-based labels, and we could not confirm how shopping, tourism and hospital trips are assigned. The Korean/foreign split also depends on the type of telecom subscription. For these reasons, successful label recovery means no more than “the provider’s classification was recovered from flows alone”.

3.2 Nodes and preprocessing

The nodes are 426 administrative districts (*dong*) of Seoul (as of December 2024), 54 cities and counties (*si/gun/gu*) of Gyeonggi and Incheon, and one “rest of country” node, for 481 nodes in total. Flows between two nodes outside Seoul (56.6% of all flows) are outside our scope and are removed; flows that involve Seoul remain (43.4% of all 2024 flows). We split days into weekdays (excluding weekends and public holidays) and weekends/holidays, use January–September (184 weekdays, 90 weekend days) for training and October–December (62 weekdays, 30 weekend days) for validation. The target of the model is the mean daily flow tensor of each period, $Y_{c,h,o,d}$ (c : day type, h : hour, o, d : origin and destination node). In the training period, Seoul-related daily flow is about 26.34 million trips on weekdays and about 21.08 million on weekends.

District codes. District codes change during the period. In 2023-02, Irwon 2-dong (11680740) in Gangnam-gu was abolished and Gaepo 3-dong (11680675) was created; in 2025-08, Yongsin-dong (11230536) in Dongdaemun-gu was split into Sinseol-dong (11230515) and Yongdu-dong (11230533). We merged each case into one node. In 2023, Bucheon is recorded under a single code (41190000), which sends 2.6% of all flows to the rest-of-country node, so in the 2023 aggregation we merged it into the Wonmi-gu node. Also, from September 2025 the file encoding changed from cp949 to UTF-8, so at first the Korean/foreign field of this period (122 days) could not be parsed and its flows were dropped entirely. We fixed the code to detect the encoding automatically and re-aggregated.

3.3 Sensors for external validation

We use two series whose measurement differs from the mobility data: Seoul-wide hourly subway entries and exits (smart card), and road volume (sum) and speed (mean) from TOPIS loop detectors. Both are summed to daily values. Bus boardings are excluded because the source gives only monthly hourly profiles, with no day-to-day variation.

3.4 Node locations

We had no district boundary polygons, so node locations on the maps are approximate. Of the 426 Seoul districts, 360 are placed at the centroid of the legal-district (*beopjeong-dong*) polygon whose name matches, and the remaining 66 are scattered deterministically within their borough (*gu*). City and county nodes use a representative point of their polygon. These locations are used only for figures, never in the analysis.

4 Method

4.1 Latent motive model

Each motive $k = 1, \dots, K$ has an origin distribution A_k and a destination distribution B_k (softmax over nodes), a daily rhythm $T_{k,c}$ for each day type (softmax over 24 hours), an intensity $s_{k,c} > 0$, and a distance scale $L_k > 0$. The expected flow of cell (c, h, o, d) is

$$\mu_{c,h,o,d} = \sum_{k=1}^K s_{k,c} T_{k,c,h} A_{k,o} B_{k,d} \exp(-\text{dist}_{od}/L_k). \quad (1)$$

We minimize the Poisson negative log-likelihood $\sum(\mu - Y \log \mu)$ with respect to the observed Y . The structure parameterizes directly “people living where (A) go where (B), when (T), and how far (L)”, so the meaning of each part can be read off.

Distance. Without district boundaries, the distance between nodes is the mean trip distance in the data (weighted by travelers). Pairs never observed in the year are filled with shortest paths on the graph built from the observed distances (Floyd–Warshall).

Training. Full-batch Adam (learning rate cosine-decayed from 0.05 to 0.002), 4,000 steps, five seeds. The intensity $s_{k,c}$ is a log-scale parameter initialized so that the total flow matches. This parameterization is the result of an error found and fixed during the analysis (Appendix A). Each model trains within seconds on a single GPU.

4.2 Baselines

- **B1: static OD $\times$ common time curve.** $\mu = \text{OD}_c[o, d] \cdot P(h \mid c)$. With 462,770 parameters it memorizes the size of every OD pair. It cannot represent reversals of direction over the day (A $\rightarrow$ B in the morning, B $\rightarrow$ A in the evening).
- **B2: shared-rhythm model.** The same model, but all motives share one daily rhythm T_c . This tests whether motives need different rhythms.
- $K = 1$ **gravity model.** A single motive.

4.3 Evaluation design: hypotheses, expectations, reasons

Table 1 lists the hypotheses and expectations fixed before training. The numbers were informed guesses.

For model selection we planned to use (i) the validation Poisson deviance (relative to total flow) and (ii) reproducibility across seeds (Hungarian-matched cosine of 0.9 or more). The reproducibility criterion was not met (Section 5.1).

5 Results

5.1 Model fit and choice of K

Six motives reach the same validation deviance as the static model (0.298 vs. 0.299) with 76 times fewer parameters (Table 2, Figure 1). At $K = 8$ the deviance is 12% lower (0.263) with 57 times fewer parameters. The shared-rhythm variant is 47% worse at $K = 8$ (0.387) and 75% worse at $K = 16$, which confirms that each motive needs its own daily rhythm. 4,000 steps is

	Hypothesis and prior expectation	Reason
H1	Motives recover the held-out purpose labels. Work and return-home motives reach purity ≥ 0.7 ; shopping, tourism and hospital do not separate. Purpose accuracy: time only 55% < motives 65–70% < supervised upper bound 75%.	The provider’s purpose rule is about trips between home, work and other places, and the model’s origin and destination distributions express the same structure. Time alone cannot tell an 8 a.m. commute from an 8 a.m. “other” trip.
H2	A motive exists whose short-term-foreigner share is at least twice the average, arriving in Myeong-dong, Jongno, Itaewon and Hongdae, with no morning peak.	Tourists have no fixed home or workplace. Nationality comes from subscription data, so it is not a label derived from movement patterns.
H3	The daily intensity of commuting motives correlates with subway entries/exits at 0.5 or more; non-commuting motives correlate weakly.	Public transit has a high share of commuting in Seoul. Telecom and smart-card measurements are independent.
H4	On holidays, commuting motives fall to 30% of normal or less; on the fireworks and rally days, arrivals in Yeouido rise without being explained; on snow days, commuting falls or is delayed.	Motives differ in elasticity. With the components fixed, one-off gatherings remain in the residual.
H5	Commuting motives have the longest distance scales, school and local motives the shortest. Return home splits into long- and short-distance motives.	People accept long distances for work and handle errands nearby. Median distance is asymmetric between work trips (4.4 km) and return-home trips (2.1 km).
H6	The top six motives by flow stay at cosine ≥ 0.8 across years.	Residential and business locations in the city change little within a year.

Table 1: Prior hypotheses and expectations.

short of full convergence. Training five times longer (20,000 steps) lowers the validation deviance by a further 3–6%, from 0.263 to 0.255 at $K = 8$ and from 0.200 to 0.188 at $K = 16$. The baseline comparison above is therefore conservative; the H1–H6 analyses below use the 4,000-step fits, and their details could change slightly with longer training.

The deviance falls smoothly with K without a clear elbow (the largest drop is from $K = 4$ to 8, then it flattens). The deviance alone therefore cannot fix K , and we chose $K = 8$ (overview) and $K = 16$ (detail) for interpretability.

Seed reproducibility falls short of the criterion. The same overall structure is broadly reproduced, but the boundaries of individual motives merge and split depending on the seed (Table 3). Individual motives should be read as examples; aggregate measures such as deviance, predictive power and sensor correlations are more reliable.

5.2 The motives found

The largest motives have natural interpretations (Table 4, Figures 2 and 3). The evening-return and morning-commute motives connect the same central business districts with the same Gyeonggi cities, in opposite directions and at opposite times. The long-trips-and-gateways motive (distance scale 25.2 km, mean OD distance 42 km) centers on long-distance hubs such as Incheon Airport, Gimpo Airport and the Express Bus Terminal, and its weekend intensity is 1.32 times its weekday intensity. The remaining small motives are close to local clusters around one or two places, such as Sinchon–Seogyo or Yeoksam–Hwayang (distance scales 0.2–1.2 km).

Model	Parameters	Val. deviance (best / mean)	Shared rhythm
Static OD $\times$ common time curve (B1)	462,770	0.299	
Motives $K = 1$ (gravity)	1,013	2.004 / 2.005	—
$K = 2$	2,026	0.596 / 0.597	0.626
$K = 4$	4,052	0.385 / 0.387	0.448
$K = 6$	6,078	0.298 / 0.310	0.409
$K = 8$	8,104	0.263 / 0.268	0.387
$K = 12$	12,156	0.221 / 0.226	0.364
$K = 16$	16,208	0.200 / 0.205	0.350
$K = 32$	32,416	0.148 / 0.158	0.322
$K = 64$	64,832	0.123 / 0.131	0.312

Table 2: Validation (October–December) Poisson deviance by number of motives; lower is better. Five seeds. The shared-rhythm variant is the best seed.

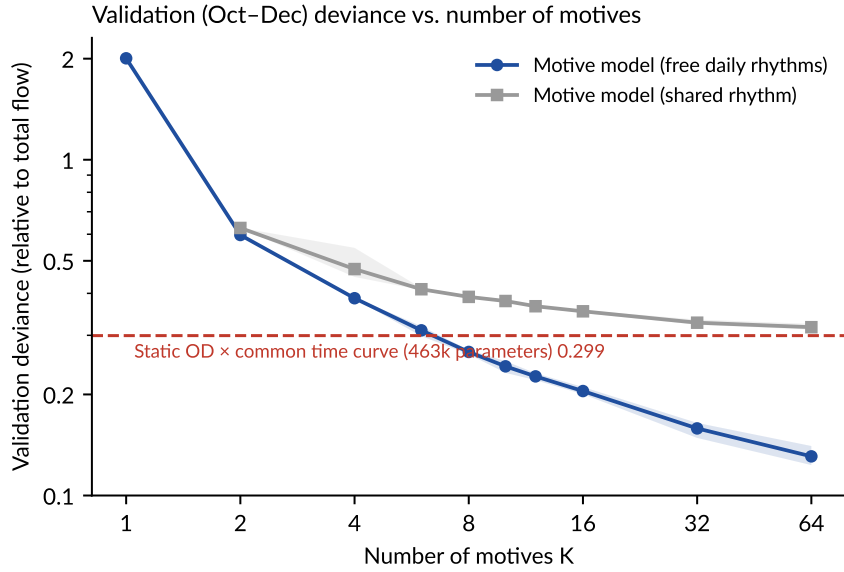


Figure 1: Validation deviance by number of motives. The shaded band is the range over seeds. The motive model catches the static baseline (dashed) from $K = 6$ and is 12% lower at $K = 8$.

5.3 H1: recovering the held-out trip purpose

The expectation was largely wrong (Table 5, Figure 4). Evaluated as designed (one purpose mix per motive), motives do worse than time alone (57.6% vs. 60.5%), because averaging a motive’s purpose mix over time discards time information. To fix this we added a variant with a purpose mix per motive and hour (motive + time). It reaches 62.1%, above time only (60.5%) and time + distance band (61.5%), but the gain is small: it closes about 36% of the gap between time only and the supervised upper bound (28% at $K = 8$). Purity is at most 0.55 for work and 0.63 and 0.56 for return home, below the expected 0.7. Shopping, tourism and hospital make up less than 3% of every motive (shopping $\leq 2.4\%$, tourism $\leq 0.7\%$, hospital $\leq 0.8\%$) and do not separate. These three account for only 0.1–0.4% of all flows and contribute almost nothing to the likelihood. This part of the expectation (no separation) held. As an interpretive limit, the purpose labels are rule-based, so even success would mean recovering the provider’s classification, not the real reasons.

K	Mean matched cosine	Flow-weighted	Share of motives above 0.9
4	0.83	0.83	0.50
6	0.80	0.81	0.38
8	0.77	0.78	0.22
12	0.76	0.79	0.31
16	0.74	0.78	0.28
32	0.71	0.74	0.14

Table 3: Reproducibility of motives across seeds (Hungarian-matched cosine of the concatenated origin, destination and daily-rhythm vectors). No K meets the 0.9 criterion.

Motive (interpretation)	Flow share	Weekday peak	Origin $\rightarrow$ destination (from actual flows)
Evening return (center $\rightarrow$ Gyeonggi)	18.2%	18 h	Yeoksam 1-dong, Yeouido-dong, Jongno $\rightarrow$ Namyangju, Goyang Deogyang-gu, Seongnam Bundang-gu
Outer-area local life	17.0%	8 h (weekend 13 h)	within Hanam, Goyang, Jingwan-dong
Morning commute (Gyeonggi $\rightarrow$ center)	12.6%	7 h	Namyangju, Bundang, Goyang $\rightarrow$ Yeoksam 1-dong, Yeouido-dong, Jongno
Long trips & gateways	12.4%	9 h (weekend 14 h)	rest of country, Gonghang-dong, Banpo 4-dong $\leftrightarrow$ Incheon Jung-gu (Incheon Airport), Banpo 4-dong
Downtown daytime	11.4%	12 h	within Jongno, Yeouido-dong, Hangangno-dong
Outer Seoul $\rightarrow$ center commute	11.2%	7 h	Segok-dong, Jingwan-dong, Doksan 1-dong $\rightarrow$ Yeouido-dong, Yeoksam 1-dong, Namyangju
Short-range evening leisure	8.8%	12 h (weekend 19 h)	within Yeoksam 1-dong, Hwayang-dong, Sageun-dong
Neighborhood nights	8.3%	8 h (weekend 21 h)	within Sinchon-dong, Seogyo-dong, Heukseok-dong

Table 4: Motives of the $K = 8$ model (seed 1, lowest training deviance). Top nodes are computed from the actual flows each motive explains, not from the parameters A, B (A_k, B_k put weight on distant nodes to compensate for distance decay).

5.4 H2: a foreigner motive

The ratio of each motive’s short-term-foreigner share to the city average (16.6%) is at most 1.25 at $K = 8$, 1.35 at $K = 16$ and 1.52 at $K = 32$ (a small motive carrying 2.6% of flow); none reaches 2. The highest is the long-distance gateway motive. The expectation was **wrong**.

Checking the cause revealed a bigger fact: short-term foreigners have almost the same hourly flow pattern as Korean nationals.

- Correlation of the hourly distributions: 0.976 on weekdays, 0.997 on weekends (long-term foreigners: 0.945 and 0.979).
- Share in the morning peak (7–9 h): 19.2% (short-term foreigners) vs. 20.6% (nationals); evening peak (17–19 h): 19.2% vs. 21.1%.
- Weekend-and-holiday to weekday flow ratio: 0.82 vs. 0.80.
- Short-term foreigners make about 4.34 million trips per day, 16.5% of the total.

This size and rhythm are impossible for tourists in the usual sense. The group is more likely one

Daily rhythm of each motive ($K = 8$)

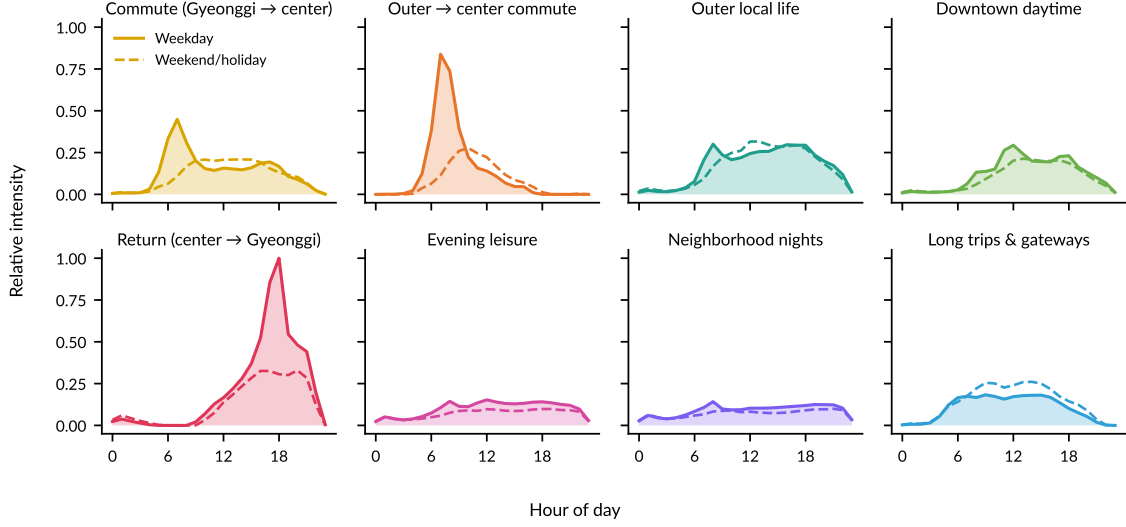


Figure 2: Daily rhythm of each motive ($K = 8$). Solid: weekdays; dashed: weekends and holidays. The evening-return and morning-commute motives have mirror-image weekday rhythms, and the long-trips motive is relatively larger on weekends.

Predictor ($K = 16$, Oct–Dec, flow-weighted)	Cross-entropy	Accuracy
Majority class (other)	—	49.0%
Motive only (design: one purpose mix per motive)	0.958	57.6%
Time only $P(\text{purpose} \mid c, h)$	0.898	60.5%
Time + distance band	0.874	61.5%
Motive + time $P(\text{purpose} \mid k, c, h)$ (added)	0.862	62.1%
Supervised upper bound $P(p \mid h, o, d)$	0.797	64.8%

Table 5: How well motives predict the held-out purpose labels. “Motive + time” is a variant added after the design (Appendix A).

with the same daily rhythm as residents (for example, short-stay workers or students classified by subscription type). The model did not fail; the label is not behaviorally distinct. We could not confirm the cause, which requires the provider’s classification criteria.

5.5 H3: agreement with independent sensors

We fix the motive components and estimate only the daily intensity $\lambda_k(\text{day})$ by Poisson EM. For each day we subtract the mean of the days within ± 28 days that share its weekday and holiday status (excluding the day itself), and compute Pearson correlations between the residuals. The commuting motives (evening return 0.77, morning commute 0.73, outer Seoul→center 0.66) move most closely with the subway, and the long-trips motive (0.16) and the outer-area local motive (0.22) the least (Figure 5). The correlation with the summed intensity of all motives is 0.59. Correlations with road volume are weaker (at most 0.53). Mean speed correlates negatively (-0.1 to -0.3), the expected direction since congestion slows traffic. Results are the same with or without public holidays. The expectation **held**: commuting motives correlate at 0.5 or more, non-commuting motives are weaker, and the long-distance motive being the least related to the subway also fits the interpretation. Since telecom and smart-card records are measured

Modeled flows by time of day (rest of country omitted); line color = dominant motive

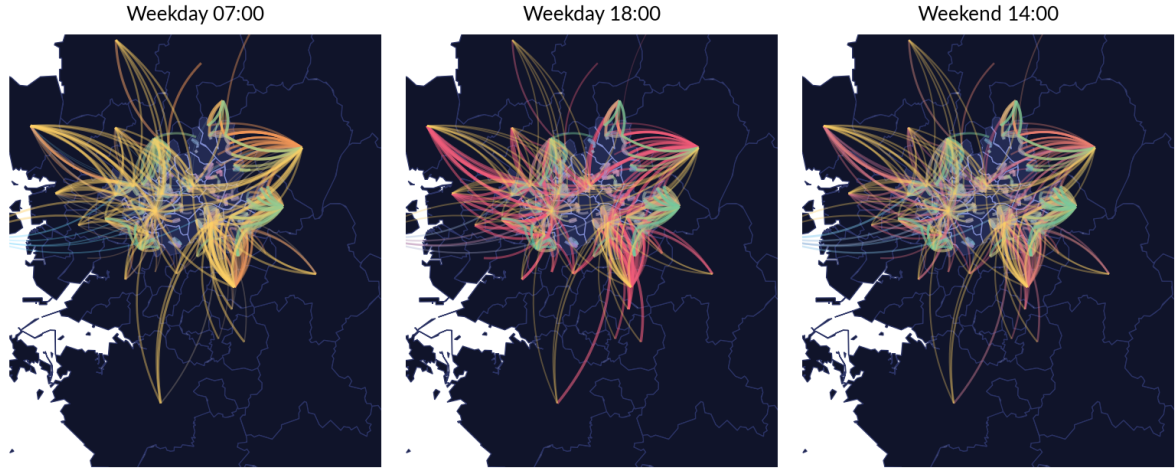


Figure 3: Flows explained by the model. At 07:00 on weekdays, gold and orange (commuting) flow toward the center; at 18:00, red (evening return) flows outward. Node locations are approximate.

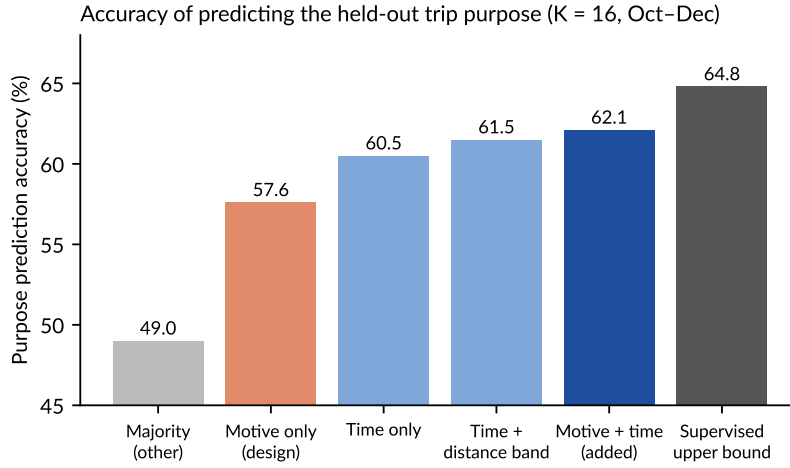


Figure 4: Accuracy of predicting the held-out purpose. Adding motives on top of time gives a small gain but stays far below the supervised upper bound.

independently, the fact that they rise and fall together on the same days is evidence that the motives reflect real behavior rather than measurement noise.

5.6 H4: event days

Holidays. On Lunar New Year (Feb 10) and Chuseok (Sep 17), the evening-return motive drops to 0.33/0.35 of a normal weekday and the outer Seoul→center motive to 0.52/0.50, but the Gyeonggi→center commute motive stays almost unchanged at 0.99/0.95 (Figure 6b). We suspect that this motive absorbs holiday visits to the center, but we did not check. The long-trips motive increases 1.85/1.59 times. The expected “commuting motives at 30% or less” did not hold (the motives that fell went to about 1/3–1/2, and one did not fall). Only the broad picture held: part of commuting falls and long-distance travel rises.

Fireworks and rallies. Looking at excess arrivals by destination district relative to the expectation under normal intensities, the largest are on the day of the Seoul International

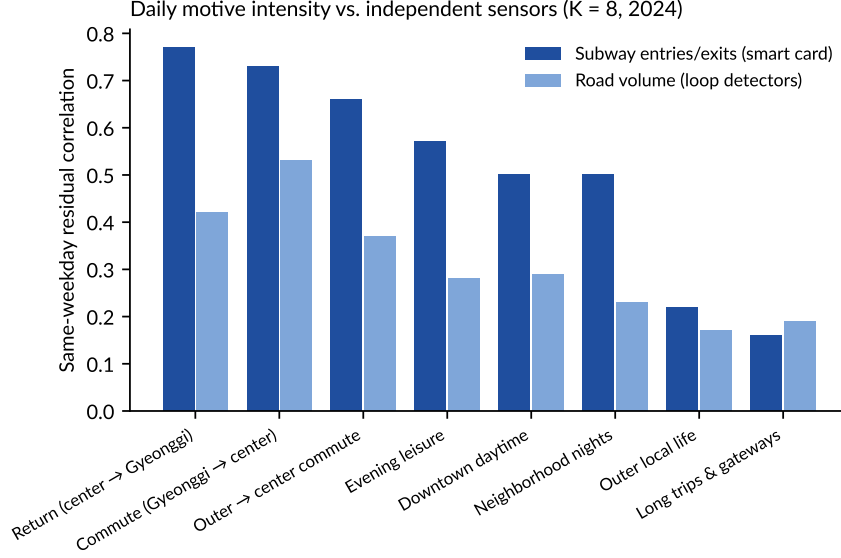


Figure 5: Same-weekday residual correlation between motive intensity and sensors. Commuting motives move most closely with the subway.

Fireworks Festival (Oct 5): Yeouido-dong +237,074 trips (+44%), and along the Han River Noryangjin 1-dong +55%, Ichon 2-dong +86% and Ichon 1-dong +58%; and on the rally days in Yeouido (Dec 7 and Dec 14): Yeouido-dong +281,452 (+49%) and +372,740 (+56%) (Figure 6a). On ordinary Saturdays (Nov 9, Dec 21), Yeouido-dong is instead at -23% and -32% . The effect of the events barely shows in the daily motive intensities (rally days $|z| \leq 1.8$) and appears only in the destination residuals. The design logic, that fixing the components pushes one-off gatherings out of the model and into the residual, held. The expectation **held**.

Heavy snow and martial law. During the heavy snow (Nov 27–28), commuting motives barely changed (0.85–1.01 of normal), and the long-trips motive fell most (0.79–0.80). The expected commuting delay was not clear. On Dec 3, the day martial law was declared, neither motive intensities nor destination residuals show a clear signal (all $|z|$ within about 2). The expectation (the event leaves a trace in daily flows) was **wrong**. Since the declaration came at night, an hourly analysis might be needed, but we did not do it.

Unexpected finding: de facto days off. On some days that are not official holidays, the commuting motives collapsed: Labor Day on May 1 (outer Seoul→center commute -44.6σ , evening return -26.5σ), the day after Memorial Day on June 7 (outer Seoul→center -11.4σ), Sep 30 (the day before the temporary holiday on Oct 1, -14.2σ), and Dec 29–31 at year end. The model finds “de facto holidays” without calendar information.

Caution. Gasan-dong in Geumcheon-gu shows -53 to -71% on every Saturday, event or control. This looks like a model-fit error rather than an event effect; we did not identify the cause. Event signals are safer to read on the positive side of the residuals.

5.7 H5: distance scales

Distance scales range widely, from 0.2km to 25km across motives ($K = 8$: 0.2, 0.3, 0.9, 1.2, 5.2, 5.5, 8.0, 25.2km). Return home splits into long- and short-distance motives as expected.

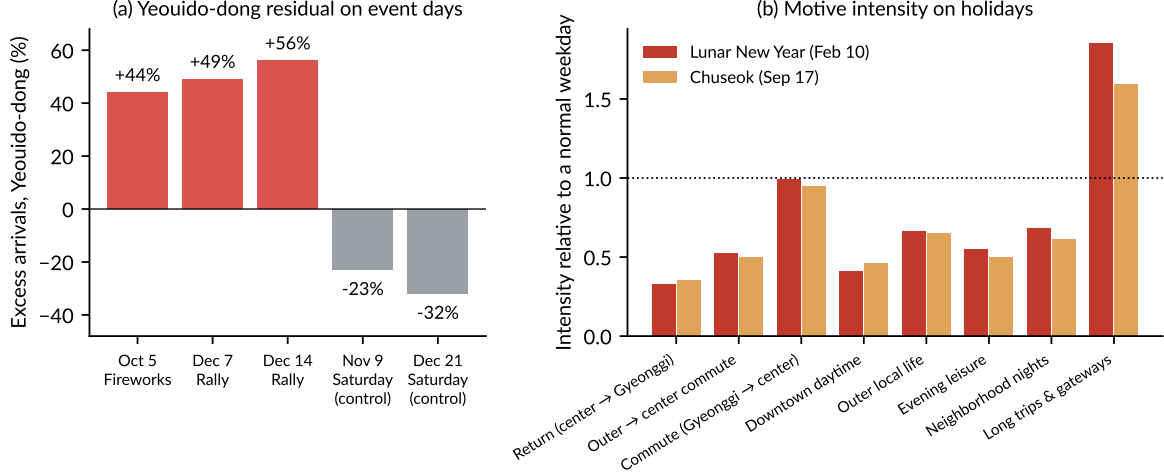


Figure 6: (a) Excess arrivals in Yeouido-dong on event days and ordinary Saturdays (observed vs. model expectation, same-weekday baseline). (b) Motive intensity on Lunar New Year and Chuseok divided by that of a normal weekday.

	2023	2025
Mean matched cosine (flow-weighted)	0.869 (0.899)	0.747 (0.754)
Reference: seeds within 2024 ($K = 8$)	0.767 (0.780)	
Deviance: 2024 components, no retraining (8k parameters)	0.261	0.241
Deviance: motive model retrained on that year	0.229	0.230
Deviance: static OD×time fitted to that year (463k)	0.267	0.258
Deviance: static OD×time estimated on 2024, applied	0.306	0.277

Table 6: Stability of motives across years. Deviance is for the mean daily flow of the whole year.

At $K = 16$, a long-distance center→Gyeonggi return (scale 7.9 km, mean OD 11.9 km) and a late-night return from Hongdae and Apgujeong to Gyeonggi and Incheon (1.8 km and 4.3 km, weekday peak at 21 h) appear separately; at $K = 32$ the return-home motives split into scales of 0.9, 3.7, 7.8 and 13.7 km. The asymmetry of median trip distance, 6.3 km for work trips and 3.9 km for return-home trips (approximated from distance-band histograms, including flows outside Seoul), fits the explanation that “return home = a mix of work→home and nearby→home”. However, **the longest scale belongs not to commuting but to airport and terminal trips** (25 km), and the commuting motives were at 7–8 km. School trips never separated at any K (school share at most 0.04). The expectation partly held.

5.8 H6: stability across years

We trained $K = 8$ independently for each year (five seeds, lowest training deviance) and compared it with the 2024 model by Hungarian matching. We also measured the deviance of the 2024 components applied without retraining to the mean daily flows of other years. 2023 behaves as expected (Table 6). Five of the eight motives are almost identical (0.97–1.00), and the flow-weighted cosine is 0.90. In 2025 the value is 0.75 and only one motive matches closely, but this is close to the value between seeds within the same year (0.77), so this method cannot tell structural change from seed noise. We withhold judgment. The stronger evidence is in the deviance. The 2024 components, used as they are, explain other years’ flows with lower deviance than a 463k-parameter static model fitted directly to that year (2023: 0.261 vs. 0.267; 2025:

	2023	2024	2025	2026
Mean commute distance (km)	11.11	11.07	10.93	10.88
Median distance (km)	7.90	7.83	7.66	7.59
Share ≥ 20 km	16.1%	16.1%	15.8%	15.8%
Mean duration (min)	48.9	49.2	48.6	48.9
Share ≥ 60 min	31.2%	31.7%	31.4%	32.2%
Seoul-bound commutes from Gyeonggi/Incheon	26.2%	26.2%	25.8%	25.7%
Mean distance, Gyeonggi/Incheon $\rightarrow$ Seoul (km)	21.0	21.0	21.0	21.0

Table 7: Work trips, January–August averages.

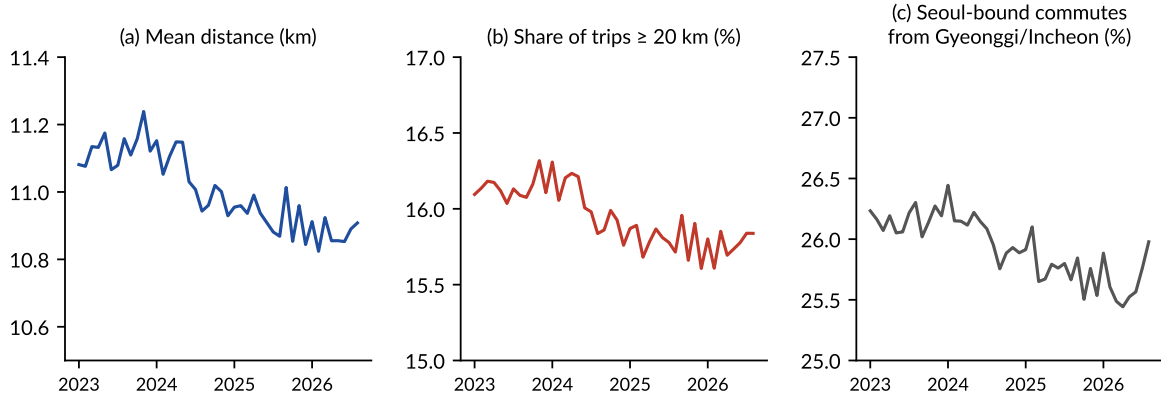


Figure 7: Monthly trends of work trips (Tuesday–Thursday medians, Korean nationals). Mean distance is falling slightly, and the long-distance share and the share starting outside Seoul are essentially flat.

0.241 vs. 0.258). The retrained and static models are in-sample values fitted to that year, while only the 2024 components are out of sample; the comparison is unfavorable to the latter, and it still wins.

5.9 Side observation: commute distances did not increase in 2023–2026

Using the same data, we checked whether commutes became longer as Seoul housing prices rose. We took work trips by Korean nationals on Tuesdays, Wednesdays and Thursdays, computed monthly medians, and removed seasonality by comparing the same months (January–August averages). Trips start or end in Seoul. After removing seasonality, the trend is -0.8% per year in mean distance and -0.9% per year in the share of trips of 20 km or more (Table 7, Figure 7). There is no evidence of an increase. Only the share of commutes of 60 minutes or more rose slightly in 2026, from 31.4% to 32.2%, but mean duration is unchanged, so this may be noise. The result should not be read as “housing prices have no effect on commute distance”. The period is short (3.6 years, and changes in commute distance due to moving may appear slowly), and the data contain no housing prices, so the relation cannot be tested. The work label is also a rule based on estimated nighttime and daytime locations. Work trips increased by 4.7% over 2023–2025, and since we cannot tell whether this is a real increase or a change in calibration, we relied on relative measures rather than counts.

5.10 Preliminary experiment: a daily autoencoder embedding

Before choosing this direction, we compressed the 24-hour daily profile of each OD pair with a variational autoencoder (16 latent dimensions), took each pair’s yearly mean as its embedding,

Feature	A→B/B→A cos.	Cluster match (years)	ARI	Origin <i>gu</i>	Dest. <i>gu</i>
Embedding (16-d)	0.203	0.816	0.624	0.121	0.122
Mean profile, PCA16	0.105	0.794	0.551	0.131	0.145
Mean profile, 48-d	0.040	0.872	0.709	0.155	0.173
Majority baseline	0	—	—	0.095	0.098

Table 8: The embedding compared with weekday/weekend mean profiles (scale-normalized). The last two columns are accuracies of predicting the origin and destination borough.

	Expectations that held	Expectations that failed
Model	Beats the static baseline at $K \approx 6$ –10. A shared rhythm is much worse. Generalizes to 2023.	Seed reproducibility of 0.9. Stability into 2025 (withheld).
H1	Shopping, tourism and hospital do not separate.	65–70% from motives alone. Purity of 0.7.
H2	—	A motive with twice the short-term-foreigner share (the label itself has a resident-like rhythm).
H3	Commuting motives correlate with the subway at ≥ 0.5 ; the long-distance motive is unrelated.	—
H4	On holidays, the evening-return and outer Seoul→center motives fall (to about 1/3–1/2) and long-distance travel rises; excess arrivals in Yeouido-dong on the fireworks and rally days.	Commuting at 30% or less on holidays (the Gyeonggi→center motive did not fall); a clear commuting delay in snow; a signal on the martial-law day.
H5	Return home splits into long and short distances; distance scales vary widely.	Commuting motives have the longest distance scale.
H6	Top motives match 2023.	—

Table 9: Prior expectations and results.

and clustered the embeddings (top 20,000 OD pairs, 2024, a “shape-only” version normalized per OD pair). The reconstruction error was far below the per-pair mean-profile baseline (0.022 vs. 0.323), and the embedding cosine of the same OD pair in 2023 was 0.855 (0.860 between the first and second halves of 2024). The embedding beat the mean profiles only in how similar round-trip OD pairs look (twice the value of a 16-d PCA), and the raw profiles were equal or better for cross-year stability and geographic information (Table 8). The embedding was, in effect, a summary of mean patterns. When daily patterns are compressed and then averaged, the structure of what moves together from day to day disappears. This is why we switched to the present approach of decomposing the whole flow into motives.

6 Discussion

6.1 Expectations versus results

Table 9 summarizes the outcome. What we did not anticipate was the resident-like rhythm of short-term foreigners, the detection of days off that are not official holidays, and the Saturday fitting error in Gasan-dong.

6.2 Interpretation

From flow counts alone, the latent motives learn a stable large-scale structure, and that structure gives meaningful signals on independent sensors and event days. The clearest result is that motives need different daily rhythms. In contrast, the motives knew little about the provider’s purpose labels beyond what time of day already tells. Either purpose is already mostly determined by time, origin and destination area, and distance, so that the model could not gain the remaining explanatory power, or the provider’s labels use information outside the structure our model sees (the character of the destination, individual histories). We could not tell which.

6.3 Limitations

1. Individual motives are sensitive to the seed (reproducibility 0.74–0.83). A few large motives are stable; small motives are close to local clusters.
2. Purpose and nationality labels are the provider’s estimates. Recovering labels is not recovering real reasons.
3. The model sees only the mean daily structure. Event effects appear only as residuals outside the components.
4. $K = 8\text{--}16$ lies in a range where the deviance is still falling, so there is no correct number of motives. Training uses 4,000 steps, short of full convergence (Section 5.1).
5. Nodes outside Seoul are coarse (city and county level), and flows between nodes outside Seoul (56.6%) were excluded.
6. Node locations are approximate, so the maps are qualitative illustrations.
7. The commute-distance trend cannot test a relation with housing prices, and the period is short.

7 Conclusion

Hourly district-level OD flows are well explained as flows produced by a few latent motives, each with a gravity-type spatial structure and its own daily rhythm. Six to eight motives match or beat a static model with 463k parameters, their intensities move together with smart-card records on the same days, and on days with known events the arrivals the model cannot explain appear exactly at those places. In contrast, these motives recover the provider’s purpose classification only slightly better than time of day alone, and the foreigner label is not behaviorally distinct. Future work is to reduce dependence on labels by adding mode-specific OD (an independent check that matches subway flows with actual entries and exits) and age data, and to refine hourly event analysis and the criteria for choosing the number of motives.

Statement on AI assistance

We used an AI assistant (Claude, Anthropic) to write analysis and figure code, to produce the figures, and to draft this English manuscript from the author’s Korean draft. All numbers reported here were computed directly from the data by that code, and none were generated by the assistant. The choice of research questions, the hypotheses, and the interpretation of the results are the responsibility of the author.

A Changes from the design and corrections

1. **Intensity-parameter error.** In the first sweep the intensity $s_{k,c}$ was a softplus parameter, but its values were in the tens of millions, so Adam with learning rate 0.05 could barely move it. The clue was that every motive had the same weekend/weekday intensity ratio as at initialization (0.80). Motive sizes were effectively fixed and small local motives were “starved of volume”; as a result, even $K = 16$ failed to beat the static baseline (0.42 vs. 0.30), which looked like hitting the predefined stopping rule. After switching to a log scale, the model wins from $K = 6$. The faulty version is kept under a separate name, and all numbers in the text are after the correction.
2. **Added H1 variant.** We added the “motive + time” predictor, which was not in the design, because the designed version discards time information and is structurally disadvantaged against the time-only baseline. The table reports both the designed version and the added variant.
3. **Nodes on motive cards.** The top origin and destination nodes were computed from the actual flows each motive explains, instead of from the parameters A_k, B_k .
4. **Event-day baseline.** At first we used a baseline mixing Saturdays and Sundays, which leaked a Saturday effect. We switched to a same-weekday baseline and added ordinary Saturdays as controls. For public holidays, same-weekday baselines are too few, so we report ratios to a normal weekday.
5. **Data.** The encoding change after 2025-09, the 2023 Bucheon code, and the splitting and replacement of district codes are described in Section 3.2. During the 2023 test, the 2024 $K = 8$ fit file was accidentally overwritten; we changed the code to save per year and retrained 2024 $K = 8$. We confirmed that the per-seed validation deviances (0.2634, etc.), the mean (0.2677) and the seed with the lowest training deviance matched the earlier values, keeping later analyses consistent.

B Reproduction

The analysis code is under `uve_experiment/od_embedding/`: `agg_labels.py` YEAR (aggregates the daily zip files), `fit_motives.py` Ks NSEED (trains models; year set by the environment variable YEAR), `eval_motives.py` K (H1, H2, H5), `daily_motives.py` K and `h3h4.py` K (H3, H4), `stability.py` and `h6.py` (reproducibility, H6), and `commute_trend.py` (commute-distance trend). A summary of results is in `RESULTS_latent_will.md` and data notes are in `DATA_NOTES.md`. Training uses PyTorch, seeds 0–4, 4,000 steps, full batch.

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